Credit EDA Assignment

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Problem Statement - 1/2

- One of the major source of income for Banking and Financial service providing company is lending loans and earning interest on the loan provided by them

- Loan providing companies lend various types of loans to consumer based on their needs. Few of them as follows
  - Housing loan
  - Car loan
  - Personal Loan
  - Business Loan

- Many a times, due to **insufficient** or **non existent credit history**, loan providing companies find it difficult to lend the loan, as some of the customers use it to their advantage by becoming a defaulter.

- When a company receives a loan application, they decide for approval of loan based on applicants' profile and come to one of the below mentioned conclusion:
  - Approve the loan application
  - Approve the loan application but with reduced amount
  - Reject the loan application

NOTE: The above mentioned conclusion falls under loan providing companies' authority to take the decision

Problem Statement - 2/2

- The loan providing companies simply cannot approve or reject the loan as there are couple of risk elements involved which are as follows:
  - Applicant fully capable of paying the loan but was denied leading to the loss of potential income which could have been received; had the loan been approved
  - Applicant unable to pay the loan but their loan application was approved leading to the loss for the company

- When a client applies for a loan, there are four types of decisions that could be taken by the client/company):
  - **Approved**: The Company has approved loan Application
  - **Cancelled**: The client cancelled the application sometime during approval. Either the client changed her/his mind about the loan or in some cases due to a higher risk of the client, he received worse pricing which he did not want.
  - **Refused**: The company had rejected the loan (because the client does not meet their requirements etc.).
  - **Unused offer**: Loan has been cancelled by the client but at different stages of the process.

- We will therefore use EDA to understand how consumer attributes and loan attributes influence the tendency to default

Approach and Methodology – 1/2

Performed EDA via the below mentioned approach

- Imported necessary libraries in Python

<u>'application_data' Dataset</u>

- Loaded the 'application_data.csv' file
- Checked the structure and metadata of 'application_data' dataset by performing the various actions like checking the shape, info of the dataset
- Started the analysis of checking the missing values by .info() function
- Kept a threshold of 40% meaning that if a column has missing value of more than 40% then that entire column was removed from the dataframe and no further analyses will be carried on those columns
- Segmenting the dataframe into Numerical and Categorical columns
- Performed Univariate Analysis on Categorical columns like NAME_CONTRACT_TYPE, CODE_GENDER, FLAG_OWN_CAR, FLAG_OWN_REALTY, NAME_INCOME_TYPE, NAME_EDUCATION_TYPE, NAME_FAMILY_STATUS, etc
- Performed Univariate Analysis on Numerical columns like AMT_INCOME_TOTAL, AMT_ANNUITY, AMT_GOODS_PRICE, AMT_CREDIT, DAYS_BIRTH, CNT_CHILDREN, DAYS_REGISTRATION
- Performed Bivariate analysis within num-num category, num-cat category and cat-cat category
- Performed Multivariate analysis

Approach and Methodology – 2/2

- Analysed the correlation between variables
- Found the top 10 correlation for the Client with payment difficulties and all other cases

- Worked on Data imbalance

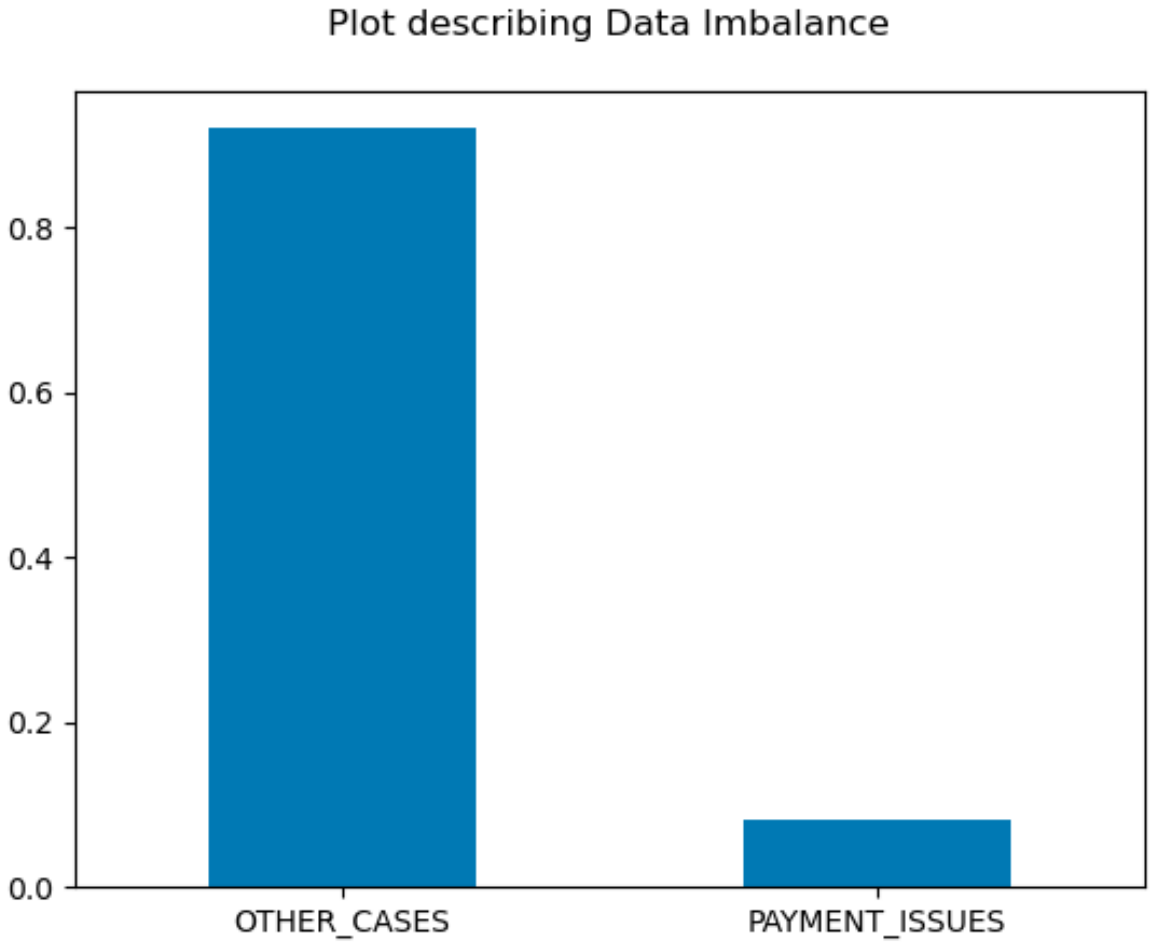
‘previous_application’ Dataset

- Loaded the ‘previous_application.csv’ file
- Checked the structure and metadata of ‘previous_application’ dataset by performing the various actions like checking the shape, info of the dataset
- Started the analysis of checking the missing values by .info() function
- Kept a threshold of 40% meaning that if a column has missing value of more than 40% then that entire column was removed from the dataframe and no further analyses will be carried on those columns
- Segmenting the dataframe into Numerical and Categorical columns
- Performed Univariate Analysis on Categorical columns like NAME_CONTRACT_TYPE, NAME_CONTRACT_STATUS, NAME_CLIENT_TYPE
- Performed Bivariate analysis within num-num category, num-cat category and cat-cat category

DATA IMBALANCE

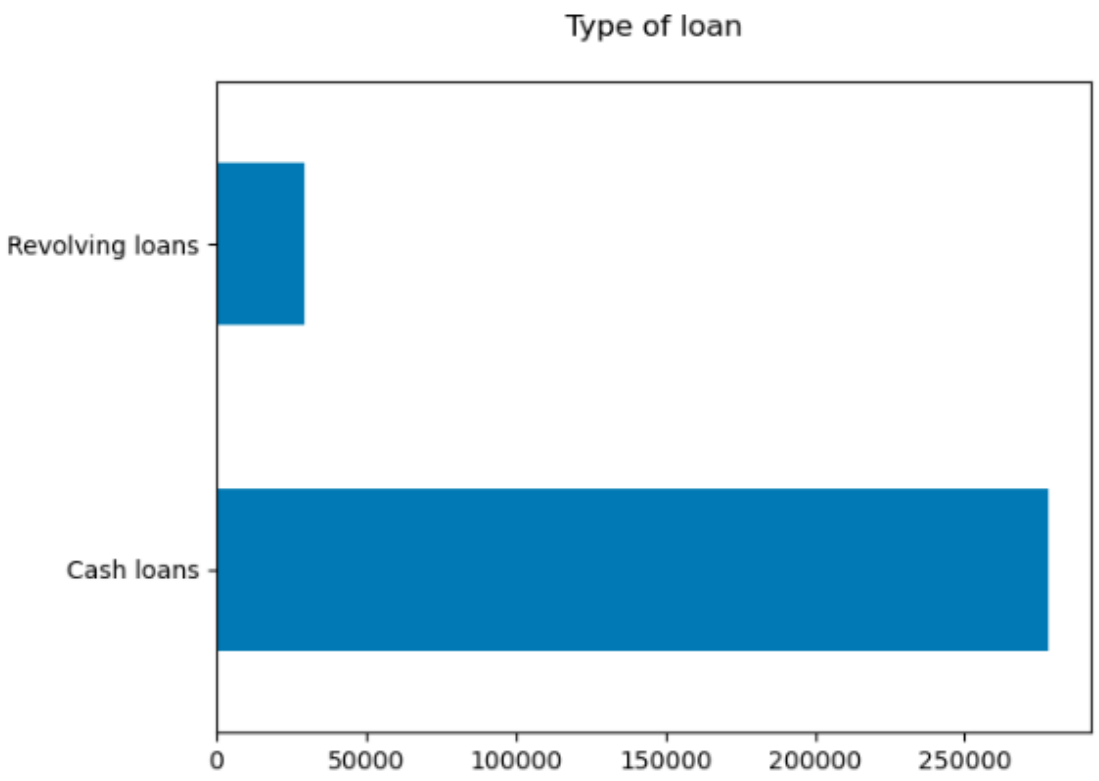
OTHER_CASES = All other cases = Non Defaulters

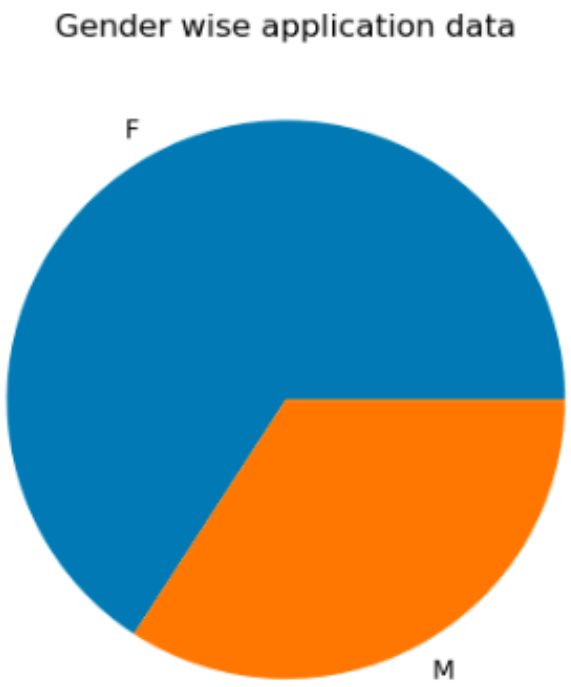
PAYMENT_ISSUES = Client with payment difficulties = Defaulters



- From the loan applications, 8% applications are such wherein clients have payment difficulties which fall under the list of **Defaulters** while rest 92% of the applicants fall in other cases which means they are **Non Defaulters**

INSIGHTS





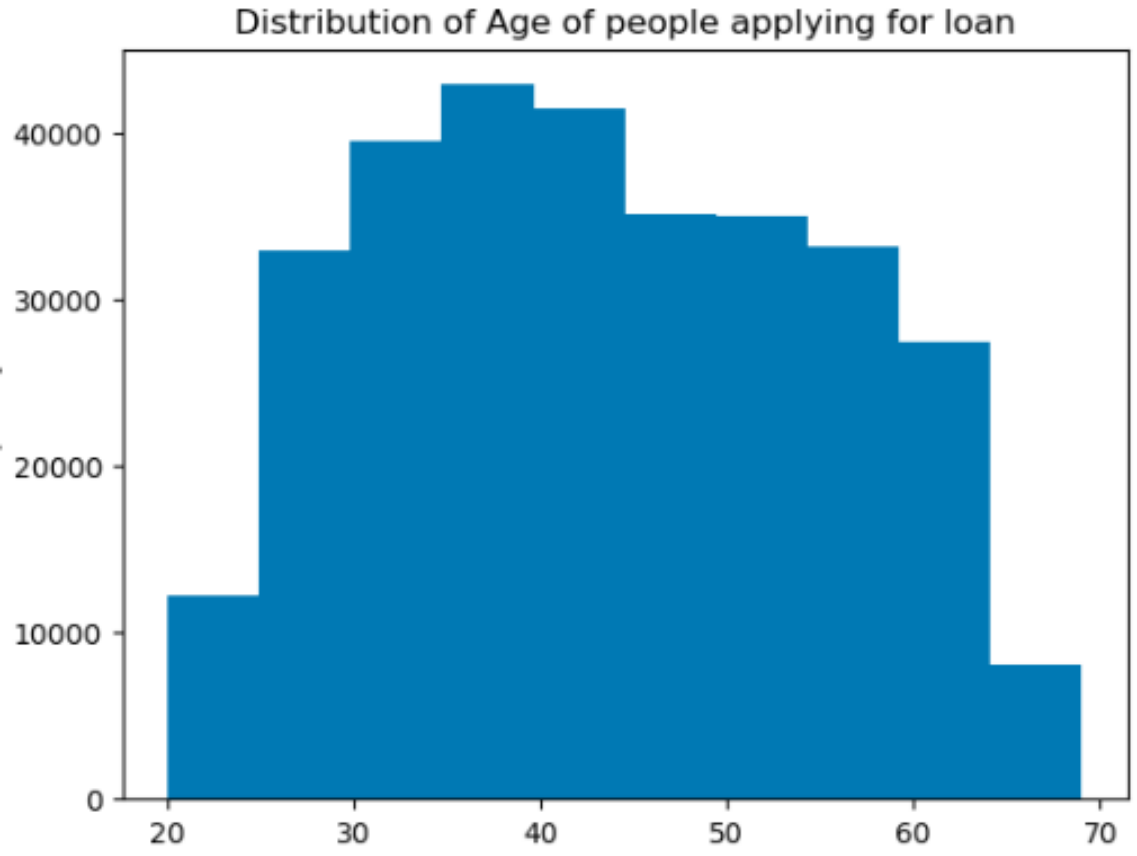
Type of loan

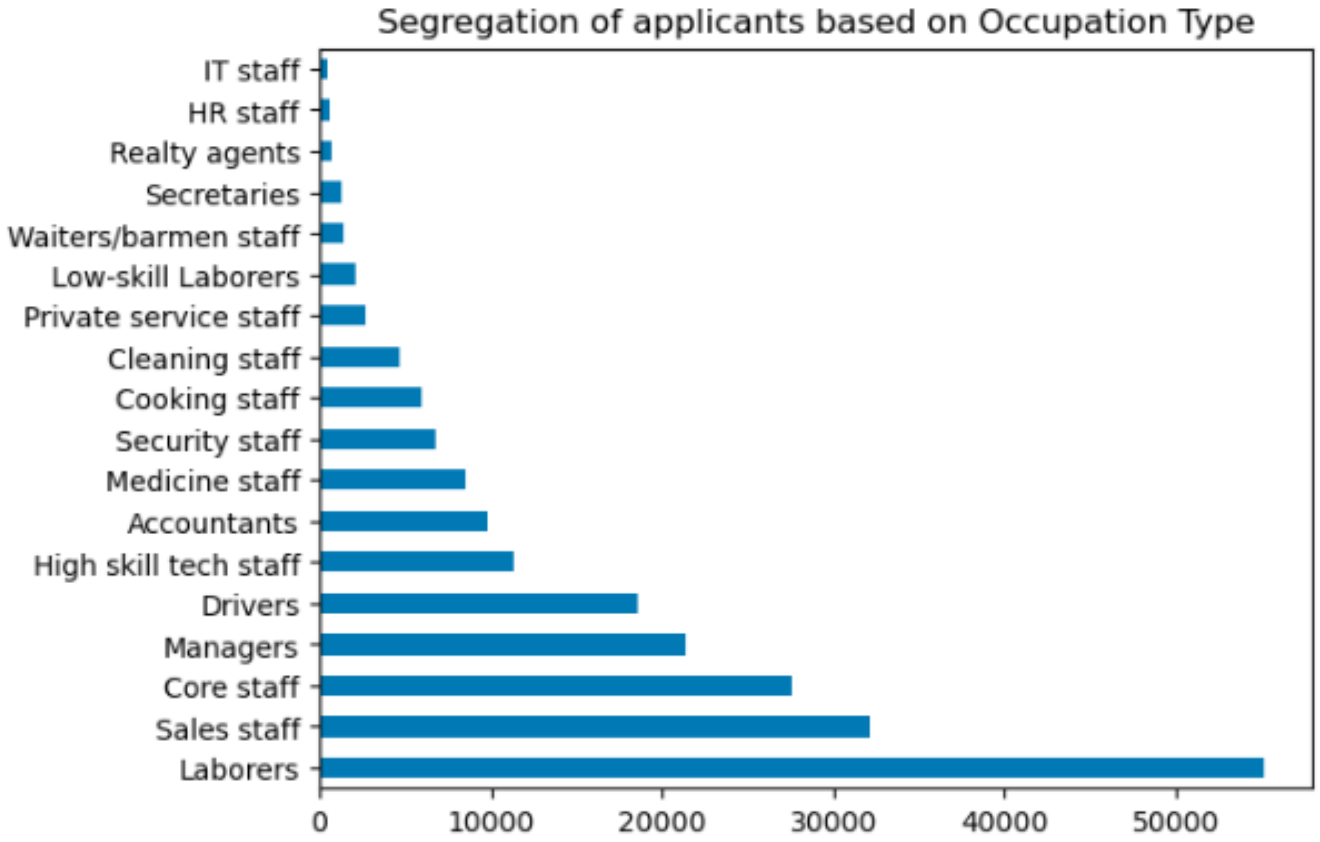
- Frequency of Cash Loan against Revolving Loan is 9:1 as 90% of the loan application have been received for Cash Loan and only 10% applications are for Revolving Loans

Segregation of applicants based on gender

- Applications received from Female: 66%
- Applications received from Males: 34%

INSIGHTS





Age Group

- Age-group of maximum of the applicants is falling in the bucket of 35-45 years

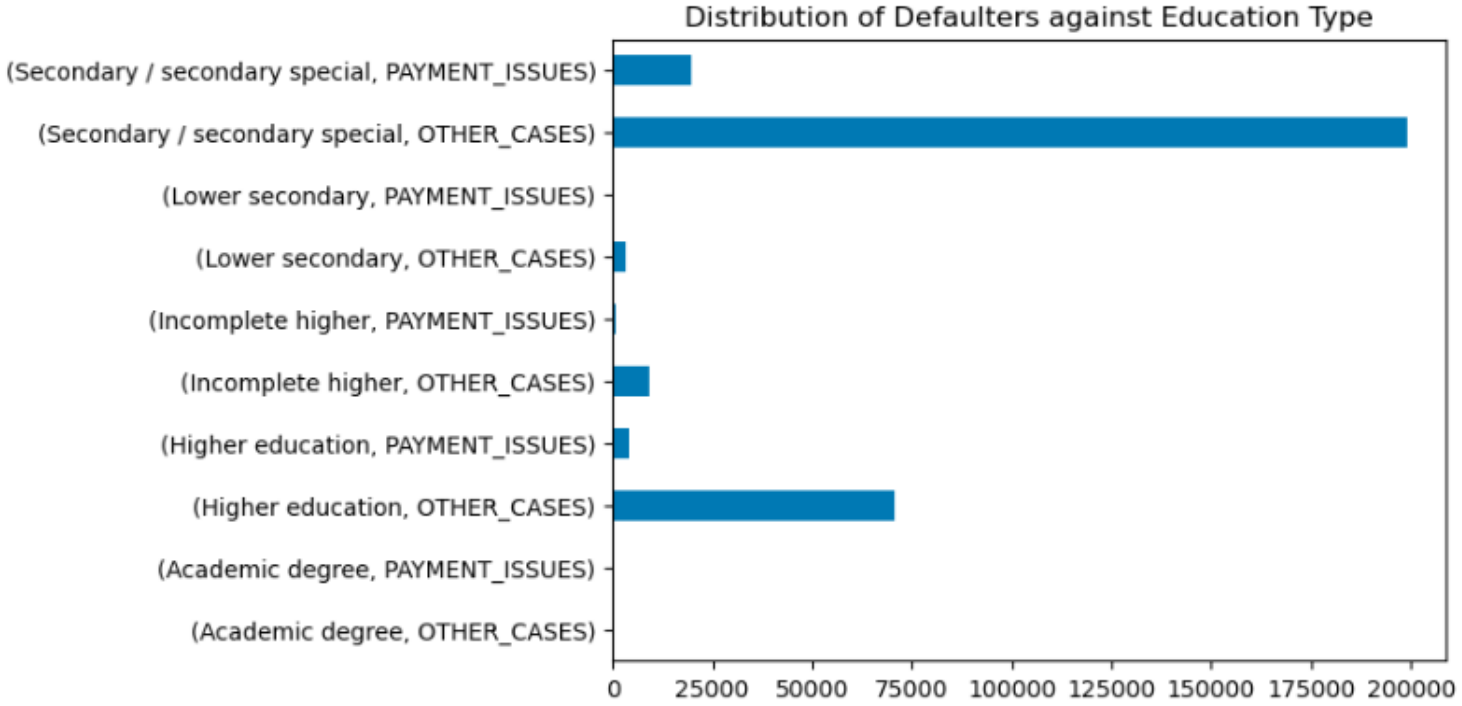
Segregation based on Occupation Type

- Top 5 applicants by Occupation Type (in %)

Laborers	26.14
Sales staff	15.21
Core staff	13.06
Managers	10.12
Drivers	8.81

INSIGHTS

Segregation of Defaulters and Non Defaulters with respect to their Education Type

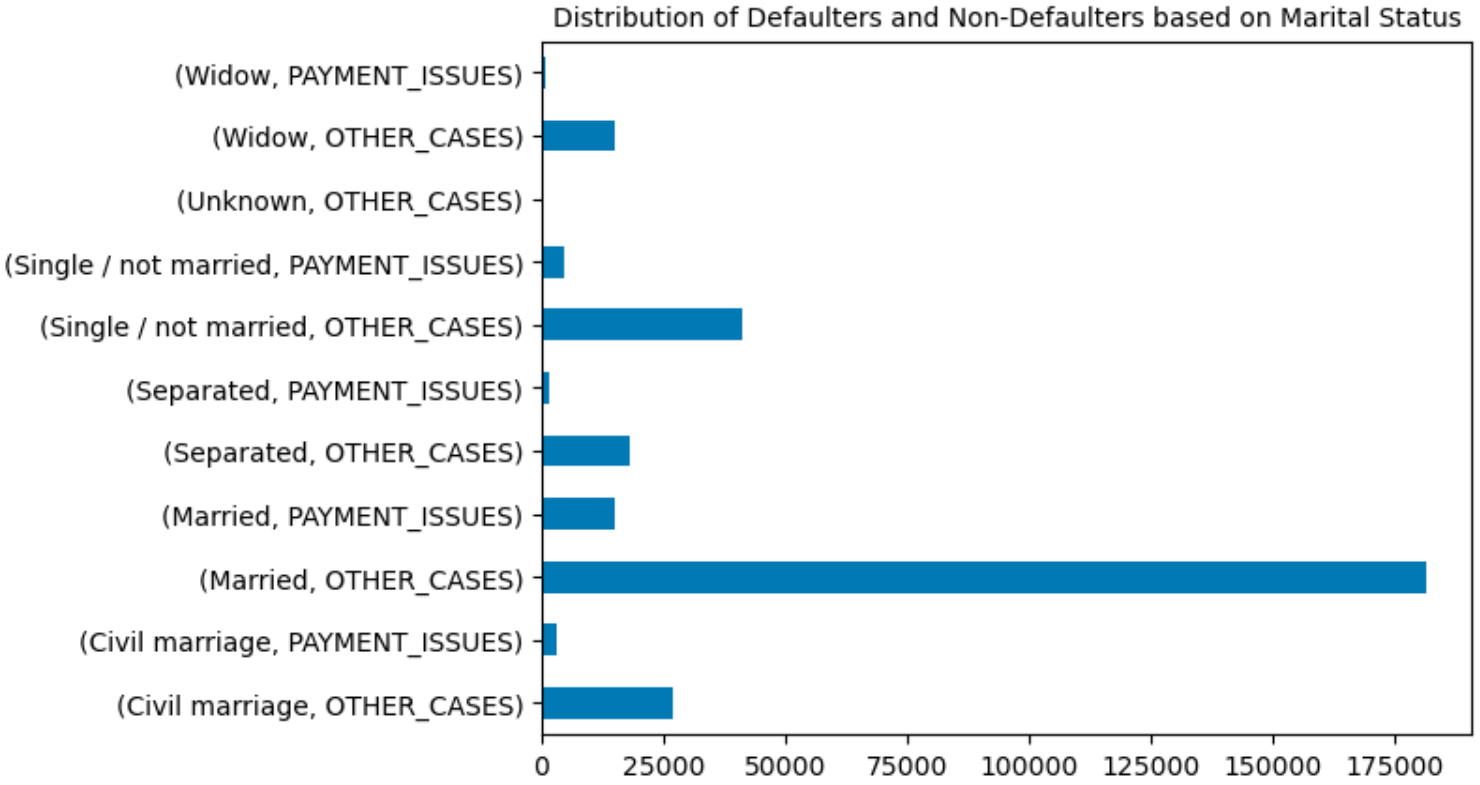


NAME_EDUCATION_TYPE	TARGET_FLAG	
Academic degree	OTHER_CASES	0.981707
	PAYMENT_ISSUES	0.018293
Higher education	OTHER_CASES	0.946449
	PAYMENT_ISSUES	0.053551
Incomplete higher	OTHER_CASES	0.915150
	PAYMENT_ISSUES	0.084850
Lower secondary	OTHER_CASES	0.890723
	PAYMENT_ISSUES	0.109277
Secondary / secondary special	OTHER_CASES	0.910605
	PAYMENT_ISSUES	0.089395

- Frequency of applicants who have completed their Secondary Education is highest which also means that frequency of defaulters is more as well as per the analysis
- Next in line are the applicants who have completed their Higher Education followed by Incomplete Higher, Lower Secondary and then Academic Degree
- Interestingly almost 10% of every Education type has Defaulters except for Higher Education which has Defaulter percentage of 5%

INSIGHTS

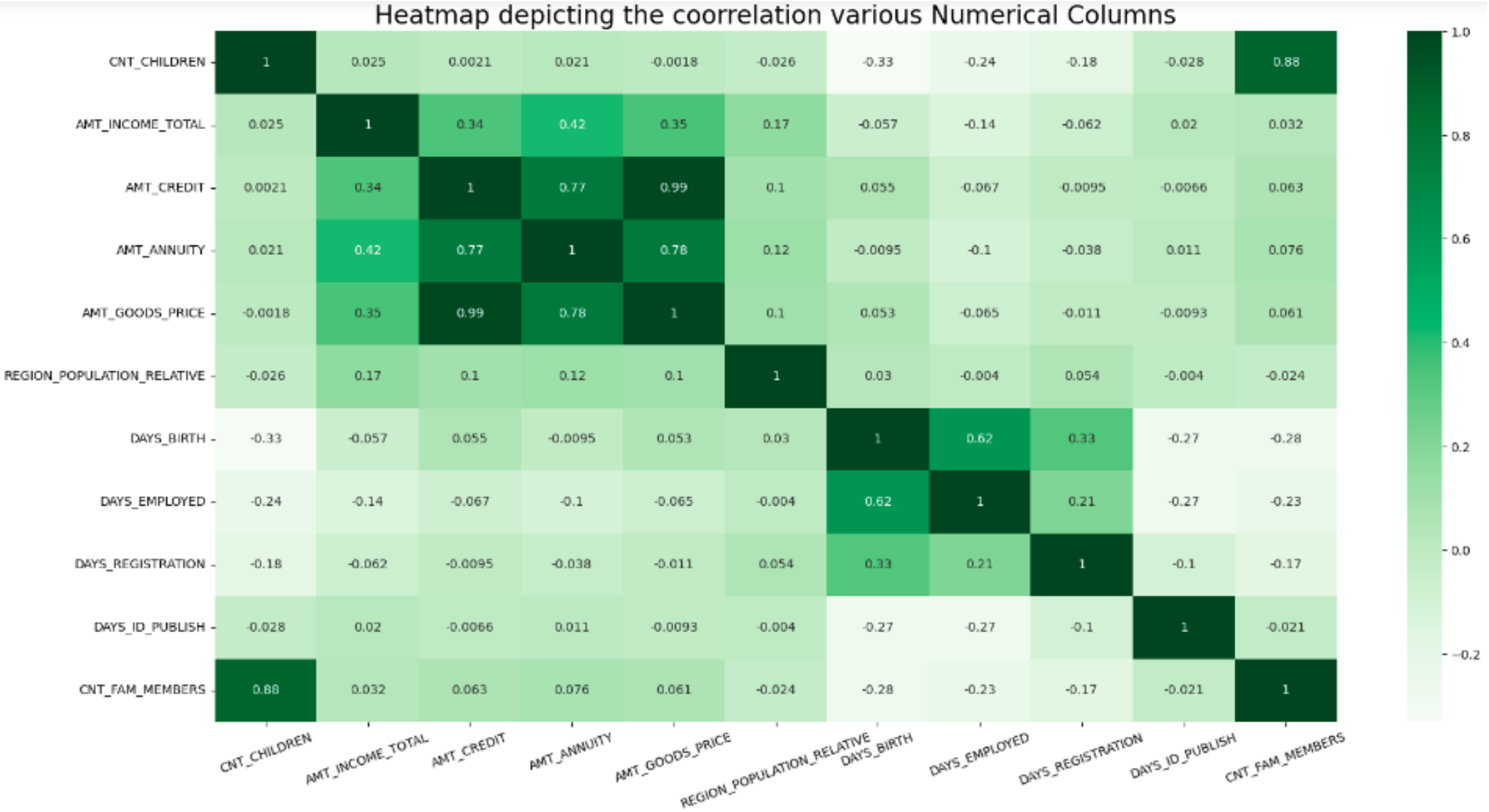
Segregation of Defaulters and Non Defaulters with respect to their Marital Status



NAME_FAMILY_STATUS	TARGET_FLAG	
Civil marriage	OTHER_CASES	90.06
	PAYMENT_ISSUES	9.94
Married	OTHER_CASES	92.44
	PAYMENT_ISSUES	7.56
Separated	OTHER_CASES	91.81
	PAYMENT_ISSUES	8.19
Single / not married	OTHER_CASES	90.19
	PAYMENT_ISSUES	9.81
Unknown	OTHER_CASES	100.00
Widow	OTHER_CASES	94.18
	PAYMENT_ISSUES	5.82

- From the data, it can be observed that applicants who are married have applied the most for loan, out of which less than 8% are Defaulters
- Applicants who are in Widow group have least applied for any loan and among them almost 6% are Defaulters which is the lowest

INSIGHTS



Heatmap depicting the correlation between Numerical Columns

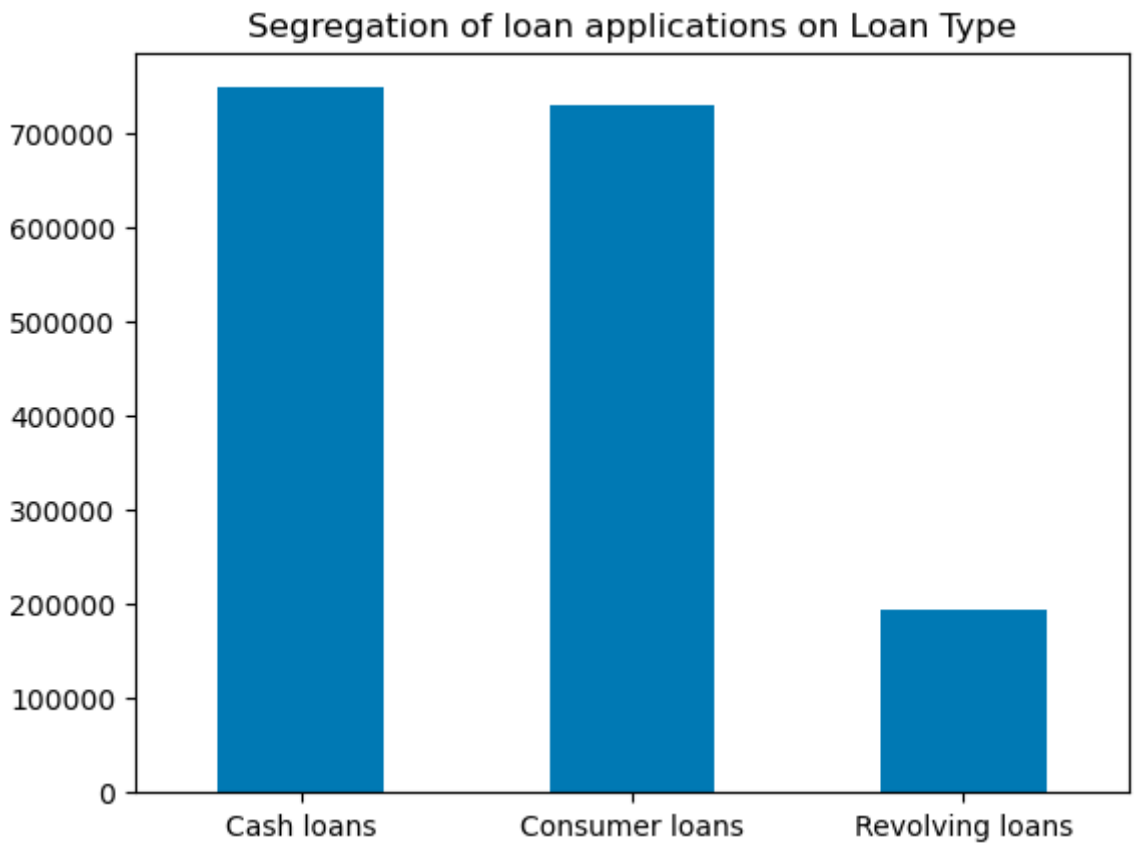
INSIGHTS

Top 10 correlation for the Client with payment difficulties & all other cases

1. AMT_CREDIT & AMT_GOODS_PRICE
2. CNT_CHILDREN & CNT_FAM_MEMBERS
3. AMT_ANNUITY & AMT_GOODS_PRICE
4. AMT_ANNUITY & AMT_CREDIT
5. DAYS_BIRTH & DAYS_EMPLOYED
6. AMT_ANNUITY & AMT_INCOME_TOTAL
7. AMT_INCOME_TOTAL & AMT_GOODS_PRICE
8. AMT_INCOME_TOTAL & AMT_CREDIT
9. DAYS_BIRTH & DAYS_REGISTRATION
10. DAYS_REGISTRATION & DAYS_EMPLOYED

INSIGHTS

Segregation of loan applications on Loan Type (Previous Application Data)



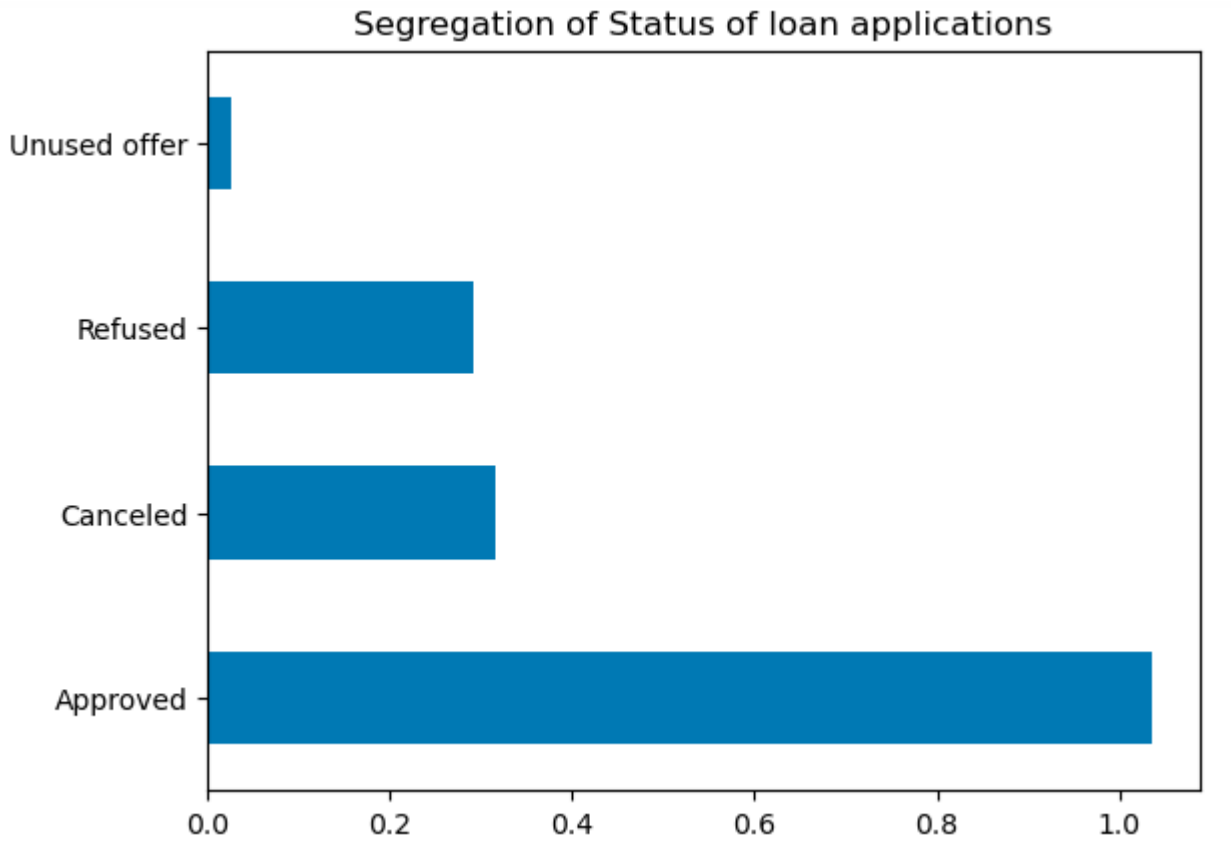
Segregation of loans (in %)

Cash loans	44.8
Consumer loans	43.7
Revolving loans	11.6

- Out of the total applications received in previously, Cash Loans and Consumer Loans dominated with 45% and 44% respectively, whereas applications received for Revolving Loans is meagre 12%

INSIGHTS

Segregation of Status of loan applications (Previous Application Data)



Segregation of status of loan application (in %)

Approved	62.1
Canceled	18.9
Refused	17.4
Unused offer	1.6

- Out of the total applications 62% of the applications were approved, 19% were cancelled either by client or company and 17% were rejected .
- There is also 1.5% applicants who cancelled during the process

Conclusion

- In application_data dataset, there were 9% of the applicants that were having payment difficulties
- Ratio of Cash Loan against Revolving Loan is 9:1
- Almost 66% of the loan applicants are females
- Age-group of maximum applicants is falling under the bucket of 35-45 years
- Applicants who are in the Occupation Type of Laborers, Sales Staff, Core Staff, Managers, Drivers accounts to almost 72% of the frequency of the total loan applications
- Frequency of applicants is most for the Education group who has completed his Secondary Education.
  - Defaulter percentage of every education group is almost 10% except for Higher Education applicants where the Defaulter percentage is 5%
- Majority of the loan applicants are married and out of them 8% are Defaulters.
  - Applicants who are widowed applied the least and 6% of them are Defaulters
- From the previous_application dataset, Cash Loans and Consumer Loans were the most with 45% and 44% respectively, whereas applications received for Revolving Loans is meagre 12%
- Out of all the loan applications, 62% of the applications were approved, 19% were cancelled either by client or company and 17% were rejected
  - There were 1.5% applications, wherein client cancelled during the process